

Training Dynamics of Overparameterized Neural Networks

A Study of Spectral Bias on S^1

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Spectral Bias

NTK on S^1

Fourier Analysis

Finite-Scale Bounds

A function is built from a coarse shape plus finer and finer detail.

target_construction_poster.png

Why does gradient descent learn simple structure first?

A spectral view of training dynamics

Thesis in one sentence

Reformulate spectral bias as an operator eigenvalue problem, then quantify how the picture changes at finite sample size and finite network width.

Three questions

- 1 What controls mode-wise learning rates?
- 2 Does the explanation survive finite data?
- 3 Does it survive finite width?

Low-frequency structure is learned before high-frequency detail.

*[Video] Spectral bias over training
one-hidden-layer MLP*

One-hidden-layer MLP

*[Video] Spectral bias over training
two-hidden-layer MLP*

Two-hidden-layer MLP

*[Video] Spectral bias over training
attention block*

Attention Block

*[Figure] Legend: colour coding
for frequency components*

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Setting: regression on S^1

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Fourier basis: $1, \cos(k\varphi), \sin(k\varphi)$.

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$$z_i = w_i^\top x + b_i$$

$$\sigma(u) = \max\{0, u\}$$

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$$\partial_t r_t = -T_t^{(m)} r_t$$

$$r_t = f_t - y$$

$T_t^{(m)}$ is the tangent operator.

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Key question: what governs the frequency-dependent decay of $\alpha_k(t) = \langle r_t, \phi_k \rangle$?

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Explicit eigenvalues

$$\lambda_k = \begin{cases} \frac{1}{4} + \frac{2}{\pi^2}, & k = 0, \\ \frac{1}{4} + \frac{1}{\pi^2}, & k = 1, \\ \frac{k^2 \pi^2 + 3}{4}, & k \geq 2 \text{ even}, \\ \frac{\pi^2 (k^2 - 1)^2}{4}, & k \geq 3 \text{ odd}. \end{cases}$$

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Interpretation

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Reference model: exact Fourier dynamics at infinite width and in the continuum.

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The ideal model is useful only if we understand its perturbations.

Perturbation I

Finite samples

$$\mu \rightsquigarrow \frac{1}{n} \sum_{i=1}^n \delta_{\varphi_i}$$

Replace the continuous measure by n random points on S^1 .

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Does the finite-width tangent operator preserve the Fourier eigenspaces and learning rates?

Both perturbations break exact diagonalization; the goal is to quantify the finite-scale error.

Main results: explicit finite-scale error bounds

On a fixed low-frequency Fourier subspace $\mathcal{H}_K$ with $d = 2K + 1$, both perturbations are controlled.

Finite samples

On the Fourier block $\mathcal{H}_K$:

$$A_{ij} = \frac{1}{n} T_{\infty}(\theta_i, \theta_j)$$

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$$\Lambda_K = P_K T_{\infty} P_K$$

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Finite samples

Thm. 5.5 + Cor. 5.6

With probability at least $1 - \delta$:

$$\|G - I\|_{\max} \leq c_1 \sqrt{\frac{\log(d/\delta)}{n}},$$

$$\|H - \Lambda^{(n)}\|_{\max} \leq c_2 \sqrt{\frac{\log(d/\delta)}{n}},$$

$$\|A\Phi - \Phi\Lambda^{(n)}\|_{\max} \leq c_3 \sqrt{\frac{\log(nd/\delta)}{n}}.$$

Frozen finite width

Thm. 5.7

With probability at least $1 - \delta$:

$$\mathbb{E} B_m^{(K)} = \Lambda_K$$

$$\|B_m^{(K)} - \Lambda_K\|_{\max} \leq c_4 \sqrt{\frac{\log(d/\delta)}{m}}$$

Summary: Fourier structure survives finite sampling and frozen finite width with explicit high-probability control.

A finite-width effect not explained by the frozen NTK

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[Figure] Training loss: frozen vs. evolving kernel — width $m = 128$

width $m = 128$: evolving kernel reaches much lower loss

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[Figure] Training loss: frozen vs. evolving kernel — width $m = 128$

width $m = 128$: evolving kernel reaches much lower loss

[Figure] Training loss: frozen vs. evolving kernel — width $m = 1024$

width $m = 1024$: frozen picture becomes competitive

- ▶ At small width, the evolving network is not well described by simply freezing the NTK at initialization.
- ▶ As width grows, the difference between frozen and evolving dynamics becomes much smaller.
- ▶ This creates the main tension: finite width can help by changing the kernel, but it also breaks the clean spectral reference model.

What changes when the operator moves?

The evolving-kernel regime keeps the same residual equation, but the operator is now time-dependent.

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Empirical picture

- ▶ Small widths show substantial kernel movement.
- ▶ Low-frequency spectral strength can increase.
- ▶ This can accelerate training loss decay.
- ▶ But eigenspaces can drift away from the Fourier reference.

Theoretical gap

The finite-sample and frozen-width regimes are controlled by concentration.

For the evolving kernel, one would need non-asymptotic control of $T_t^{(m)} - T_0^{(m)}$ over training time.

Thesis takeaway: the frozen NTK gives a controlled spectral reference; finite-width training shows where that reference becomes insufficient.

Future direction: mathematical structure in function space as an explanation for training and generalization.